NATIONAL INCOME & AGGREGATES

CBSE CLASS 12 MACROECONOMICS • COMPREHENSIVE STUDY NOTES

UNIT 1
Weightage: 10 Marks

> **Unit Overview**
> National Income is the net money value of all final goods and services produced by normal residents of a country during an accounting year. This unit evaluates macroeconomic identity, structural classifications of goods, circular money flows, and the three standardized measurement methodologies.

1. CORE CONVERSION FRAMEWORK

Every national income aggregate is built from three basic conversion pairs. Master these identity transformations to solve all numerical aggregates smoothly:

1. Gross vs. Net
Gross = Net + Depreciation
Net = Gross − Depreciation
*Depreciation = Consumption of Fixed Capital

2. Domestic vs. National
National = Domestic + NFIA
Domestic = National − NFIA
*NFIA = Factor Income from Abroad − Paid

3. Market Price vs. Factor Cost
MP = FC + NIT
FC = MP − NIT
*NIT = Net Indirect Taxes (GST − Subsidies)

2. KEY THEORETICAL CONCEPTS

A. Classification of Goods

Category	Definition & Core Criterion	Key Example
Final Goods	Crossed the boundary line of production; ready for end-use by final consumers or investors. Included in National Income.	Car bought by household, Machinery bought by firm.
Intermediate Goods	Within production boundary; used as raw material or resold in the same year. Excluded to avoid double counting.	Milk bought by a dairy shop, Flour used by a baker.
Consumer Goods	Directly satisfy human wants (Durable, Semi-durable, Non-durable, Services).	Bread, Clothes, Refrigerator, Doctor's care.
Capital Goods	Fixed assets of producers used repeatedly in production for several years; undergo wear and tear.	Tractors, Industrial Machinery, Factory buildings.

B. Stocks vs. Flows

- **Stock Variable:** Measured at a *specific point in time*. It has no time dimension. (e.g., Wealth, Capital stock, Money supply as of 31st March, Bank balance).
- **Flow Variable:** Measured over a *period of time* (per hour, per month, per year). It carries a time dimension. (e.g., Income, Expenditure, Capital formation, Interest earned).

C. Circular Flow of Income (Two-Sector Economy)

In a simple closed economy with two sectors (Households and Firms) without government or foreign sector:

- **Real Flow:** Flow of factor services (Land, Labour, Capital, Enterprise) from Households to Firms, and corresponding flow of goods and services from Firms to Households.
- **Money Flow:** Flow of factor payments (Rent, Wages, Interest, Profit) from Firms to Households, and spending of that money on purchasing final goods from Firms.
- **Equilibrium Identity:** $\text{Total Production} \equiv \text{Total Factor Income} \equiv \text{Total

3. THE THREE MEASUREMENT METHODS

METHOD 1: VALUE ADDED (PRODUCT) METHOD

Measures the contribution of each producing enterprise in the domestic territory.

1. Value of Output = Sales + Change in Stock (ΔS)

[Note: ΔS = Closing Stock − Opening Stock]

2. Gross Value Added at Market Price (GVA_{mp}) = Value of Output − Intermediate Consumption

3. Σ GVA_{mp} = GDP_{mp}

Precautions: Do not include sale of second-hand goods (value already counted). Include imputed value of self-consumed output. Exclude intermediate consumption to prevent double counting.

METHOD 2: INCOME METHOD

Measures total factor income generated by factors of production inside domestic territory.

NDP_{fc} (Net Domestic Income) = Compensation of Employees + Operating Surplus + Mixed Income

- **Compensation of Employees (COE):** Wages/Salaries in cash + Wages in kind + Employer's contribution to Social Security Schemes + Retirement Pension.
- **Operating Surplus:** Income from property (Rent + Royalty + Interest) + Income from entrepreneurship (Profit).
  *Profit = Corporate Tax + Dividend + Retained Earnings (Undistributed Profit).
- **Mixed Income:** Income of self-employed individuals using their own labor, capital, and land.

Precautions: Exclude Transfer Payments (pensions, scholarships, pocket money). Exclude windfall gains (lotteries, capital gains). Include imputed rent of self-occupied house.

METHOD 3: EXPENDITURE METHOD

Measures aggregate final expenditure on goods and services produced within domestic territory.

GDP_{mp} = PFCE + GFCE + GDCF + Net Exports (X − M)

- **PFCE:** Private Final Consumption Expenditure by households and private non-profit institutions.
- **GFCE:** Government Final Consumption Expenditure on public welfare services.
- **GDCF (Gross Domestic Capital Formation):** Gross Fixed Capital Formation + Change in Stock (ΔS).
- **Net Exports (X − M):** Exports (X) − Imports (M).

Precautions: Exclude expenditure on second-hand goods and financial assets (shares, bonds). Include self-consumed goods investment expenditure.

4. REAL GDP, NOMINAL GDP & WELFARE

Indicator	Formula & Mechanism	Economic Significance
Nominal GDP	GDP at Current Prices = Quantity (Q) × Current Price (P_1) Measured using current year market prices.	Can rise due to mere inflation without any physical increase in output.
Real GDP	GDP at Constant Prices = Quantity (Q) × Base Price (P_0) Measured using base year prices.	True indicator of physical output growth and economic performance.
GDP Deflator	GDP Deflator = (Nominal GDP / Real GDP) × 100	Measures overall price level change (inflation index).

Limitations of GDP as a Welfare Indicator

5. SOLVED MASTER NUMERICAL

Illustration: Calculate (a) Net Domestic Product at Factor Cost (NDP_{fc}) and (b) National Income (NNP_{fc}) from the given data (in ₹ Crores):

1. Compensation of Employees = 800
2. Operating Surplus = 600
3. Mixed Income of Self-Employed = 400
4. Net Factor Income to Abroad = 50
5. Net Indirect Taxes = 120
6. Consumption of Fixed Capital = 80

Solution:

(a) Calculation of Domestic Income (NDP_{fc}):

Using Income Method: NDP_{fc} = Compensation of Employees + Operating Surplus + Mixed Income

NDP_{fc} = 800 + 600 + 400 = ₹ 1,800 Crores

(b) Calculation of National Income (NNP_{fc}):

$NNP_{fc} = NDP_{fc} + NFIA$

Note: Net Factor Income *To* Abroad is given as 50. Therefore, NFIA = −50.

NNP_{fc} = 1800 + (−50) = ₹ 1,750 Crores

6. IMPORTANT EXAM QUESTIONS

Conceptual **Q1. How should 'Intermediate Goods' be treated while estimating National Income? Why?** 3 Marks

Answer: Intermediate goods must be **excluded** from the estimation of National Income. *Reason:* Their value is already included in the value of final goods. Including intermediate goods separately would lead to **double counting**, artificially inflating the true national output.

Distinction **Q2. Distinguish between Factor Payments and Transfer Payments with examples.** 3 Marks

Answer:

- **Factor Payments:** Payments made to factors of production for rendering productive factor services (e.g., Rent, Wages, Interest, Profit). These are earned concepts and included in National Income.
- **Transfer Payments:** Unilateral payments received without rendering any productive service in return (e.g., Old-age pensions, Scholarships, Unemployment allowance). These are unearned concepts and excluded from National Income.

Application **Q3. "Real GDP is a better indicator of economic growth than Nominal GDP." Justify.** 4 Marks

Answer: Nominal GDP is calculated at current year prices, so it can increase even if physical production remains constant simply due to inflation. Real GDP is measured at constant base-year prices and changes *only* when physical output increases. Thus, Real GDP reflects true economic growth and real availability of goods/services.

HOTS **Q4. Explain how 'Externalities' act as a limitation of GDP as an index of welfare.** 4 Marks

Answer: Externalities refer to benefits or harms caused by a firm or individual for which they are neither paid nor penalized.

- **Negative Externalities:** Pollution generated by factories harms public health, reducing welfare, but GDP does not deduct this cost. Thus GDP overstates welfare.
- **Positive Externalities:** A beautiful private garden created by an owner pleases passers-by, increasing welfare, but GDP ignores it. Thus GDP understates welfare.

7. HIGH-YIELD PRACTICE QUESTION BANK

Categorization Practice: Included or Excluded from National Income?

Item / Economic Transaction	Status	Economic Reason
1. Purchase of car by a taxi driver	**INCLUDED**	It is a investment spending / capital purchase for production.
2. Financial help given to flood victims	**EXCLUDED**	Transfer payment (unilateral payment, no production).
3. Imputed rent of self-occupied house	**INCLUDED**	It represents factor service provided by owned capital.
4. Purchase of shares/bonds of a company	**EXCLUDED**	Mere paper claim; does not directly produce goods/services.
5. Commission paid to broker for 2nd-hand car	**INCLUDED**	Payment for productive service rendered by broker this year.

Short & Long Answer Questions for Exam Revision

Theory **Q5. Differentiate between Capital Loss and Depreciation.** **3 Marks**

Answer: Depreciation refers to expected wear and tear, normal obsolescence, and routine damage of fixed assets during production (covered by reserve funds). **Capital Loss** refers to unforeseen destruction of assets due to natural calamities or fires (covered by insurance).

Assertion-Reason **Q6. Analyze Assertion (A) & Reason (R):** **3 Marks**

Assertion (A): GDP at Factor Cost (GDP_{fc}) can be greater than GDP at Market Price (GDP_{mp}).
Reason (R): When Subsidies exceed Indirect Taxes, Net Indirect Taxes (NIT) become negative.
Answer: Both (A) and (R) are true, and (R) is the correct explanation. Since $GDP_{mp} = GDP_{fc} + NIT$, if subsidies > indirect taxes, NIT is negative, making $GDP_{fc} > GDP_{mp}$.

Numerical Practice **Q7. Calculate GDP_{mp} using Value Added Method from the given firm data:** **4 Marks**

Data: Sales = ₹ 800 Lakhs, Opening Stock = ₹ 40 Lakhs, Closing Stock = ₹ 20 Lakhs, Intermediate Purchases = ₹ 300 Lakhs, Subsidy = ₹ 10 Lakhs.
Solution: Value of Output = Sales + (Closing Stock − Opening Stock) = 800 + (20 − 40) = ₹ 780 Lakhs.
GVA_{mp} (GDP_{mp}) = Value of Output − Intermediate Consumption = 780 − 300 = **₹ 480 Lakhs**.

Conceptual **Q8. Are Corporate Tax and Retained Earnings added separately to Operating Surplus while calculating Domestic Income?** **3 Marks**

Answer: No. Corporate Tax and Retained Earnings (Undistributed Profit) are sub-components of Profit. Operating Surplus already includes total Profit. Adding them separately would lead to double counting.

Theory **Q9. Explain the relationship between Gross Value Added at Factor Cost (GVA_{fc}) and Net Domestic Product at Factor Cost (NDP_{fc}).** **4 Marks**

Answer: Gross Value Added at Factor Cost (GVA_{fc}) is the value added by an individual producing unit after deducting intermediate purchases and net indirect taxes. When GVA_{fc} of all producing enterprises across Primary, Secondary, and Tertiary sectors in the domestic territory are summed up, we get Gross Domestic Product at Factor Cost (GDP_{fc}). Deducting depreciation gives Net Domestic Product at Factor Cost ($NDP_{fc} = \Sigma GVA_{fc} - Depreciation$).

> **Exam Tips for 100% Accuracy in Numericals**